

Input-Output Multipliers: The Least Wrong Way

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Introduction to Multipliers

To understand how MASS culture looks to model economic impact we must first understand the Input-Output (I-O) Model and its multipliers.

We will start with a simple example. “Simon’s Dance Academy” has been awarded a **\$500,000 grant**. Using Input-Output Model multipliers we want to know how that investment will impact the broader economy in terms of Jobs and GDP.

The model we will use was originally developed by Leontief (1936) and has since been expanded upon. We will leave the math of the model for later, but what we need to know now is that this model gives use three multipliers. These capture direct, indirect, and induced effects of a output shock in a economy. In practice there is a set of multipliers for each industry. These industries are defined by IOIC (Input Output Industry Classification) codes.

To explain the direct, indirect and induced multipliers/effects we will first classify our academy’s industry as the *Performing arts, spectator sports and related industries, and heritage institutions [BS71A000]* which has a set of three national published multipliers. Using the 2022 multipliers we have the following.

Effect	Multiplier
Direct	0.510
Indirect	0.309
Induced	0.257

Quick definitions

Output - In I-O Models means means gross output which is the total value of everything an industry produces, including intermediate goods sold to other industries, not just final sales.

Direct - The value added by the organization itself: its wages, operating surplus, and taxes on production (i.e. its own gross output minus what it spent on intermediate inputs).

Indirect - Activity generated up the supply chain. The organization buys from suppliers, who buy from their suppliers, and so on.

Induced - Activity generated when employees (at the organization and its suppliers) spend their wages on goods and services in the broader economy.

Now we apply our multipliers to output. Note that most cases this is operating revenues or expenditures, preferably the later.

Direct Effect

The academy pays its instructor and spends on costumes, marketing, and utilities. The direct effect captures the value added net of those intermediate purchases.

$$\$500,000 \times 0.510 = \$255,000$$

Indirect Effect

To operate, the academy purchases sound equipment from a local supplier, who in turn buys components from their own distributor. The indirect effect captures this chain of upstream activity.

$$\$500,000 \times 0.309 = \$154,500$$

Induced Effect

The instructor and the supplier's staff spend their wages locally on groceries, rent, and restaurants. The induced effect captures that final round of consumer spending.

$$\$500,000 \times 0.257 = \$128,500$$

Total GDP Impact

Summing all three effects gives the full economic footprint of the \$500,000 grant.

$$\$255,000 + \$154,500 + \$128,500 = \$538,000$$

Every dollar granted to the academy generates **\$1.076 of GDP** across the economy.

Applying Multipliers

- Multipliers are built to measure the effect of **new** spending added to the economy, not spending that is already happening. The baseline activity of the sector is already baked into the data used to build the multipliers [Miller and Blair, 2009].
- For a single organization or grant, this is not a problem. The shock is small, so the assumptions behind the model hold [Welsh Government, 2024].
- For an entire sector, the assumptions start to break down. The model assumes supply chains can expand proportionally no matter how big the shock is which is not realistic at sector scale [Welsh Government, 2024; West and Jackson, 1998].
- Applying multipliers at the sector level is still common practice, but the results should be treated as a rough indication of scale rather than a precise impact estimate.

Mixing Multipliers

Now that we have a idea of what the I-O Model does, we can explain some of its math and show how we can mix multipliers to accomidate organizations that do not fall cleanly into a industry.

Define the Model

The technical coefficients matrix $\mathbf{A}$ describes the supply chain of an entire economy. Each entry a_{ij} answers a simple question: *when industry j produces one dollar of output, how many cents does it spend buying inputs from industry i ?* For example, if the performing arts industry spends 15 cents per dollar of output on printing and publishing, that's one entry in $\mathbf{A}$.

The Leontief inverse $\mathbf{L} = (\mathbf{I} - \mathbf{A})^{-1}$ traces those supply-chain purchases through every round of indirect spending. It answers: *if we inject one dollar of final demand into industry j , how much total output is required from every industry in the economy — including the suppliers' suppliers, and so on?*

GDP and jobs multipliers are what Statistics Canada publishes. They are not raw column sums of $\mathbf{L}$. Instead, StatCan applies **value-added ratios** (GDP per dollar of output) and **labour coefficients** (jobs per dollar of output) to each row of $\mathbf{L}$, then sums down each column. The result: *for every dollar of expenditure in industry j , how many dollars of GDP and how many jobs are generated economy-wide?*

Symbol	Plain-language meaning
$\mathbf{x}$	Total output vector: gross production required from each industry in equilibrium
$\mathbf{y}$	Final demand vector: expenditure by households, government, exports, etc.
$\mathbf{A}$	Who buys from whom, and how much, per dollar of output
$\mathbf{L}$	The cumulative supply-chain ripple effects of a dollar of demand
$\mathbf{v}$	GDP generated per dollar of output in each industry (value-added ratio)
$\mathbf{e}$	Jobs supported per million dollars of output in each industry (labour coefficient)
$m_j^{GDP} = \mathbf{v}'\mathbf{L}_{\cdot j}$	GDP multiplier: dollars of GDP per dollar of expenditure in industry j
$m_j^{Jobs} = \mathbf{e}'\mathbf{L}_{\cdot j}$	Jobs multiplier: jobs per million dollars of expenditure in industry j
w_k	Share of an organization's activity in industry k ($\sum_k w_k = 1$)
D	Total expenditure of the organization

$$\mathbf{x} = \mathbf{A}\mathbf{x} + \mathbf{y} \implies \mathbf{x} = (\mathbf{I} - \mathbf{A})^{-1}\mathbf{y} = \mathbf{L}\mathbf{y}$$

GDP and jobs impacts of a demand shock $\mathbf{y}$:

$$\Delta GDP = \mathbf{v}'\mathbf{L}\mathbf{y} \quad \Delta Jobs = \mathbf{e}'\mathbf{L}\mathbf{y}$$

When the shock targets a single industry j with magnitude D :

$$\Delta GDP = D \cdot \underbrace{\mathbf{v}'\mathbf{L}_{\cdot j}}_{m_j^{GDP}} \quad \Delta Jobs = \frac{D}{10^6} \cdot \underbrace{\mathbf{e}'\mathbf{L}_{\cdot j}}_{m_j^{Jobs}}$$

Note that induced effects are captured by having two models; a closed and a open one. The closed model endogenizes (includes) households, so the induced effects are captured in the Leontief inverse. The open model does no such thing. When you subtract the open from the closed you get the induced effects. Essentially you add a new column and row to the $\mathbf{A}$ matrix.

$$\mathbf{A}^* = \begin{bmatrix} \mathbf{A} & \mathbf{h}_c \\ \mathbf{h}_r' & 0 \end{bmatrix}$$

Where:

- $\mathbf{A}$: industry-to-industry purchases per dollar of output
- $\mathbf{h}_c$: household spending on each industry per dollar of income
- $\mathbf{h}_r'$: wages paid to households per dollar of industry output
- 0: households are not an industry, so no self-purchase

And then re-run the model same as before.

A Toy Economy

Imagine we have a economy with three industries: **Performing Arts**, **Film/Video**, and **Publishing**.

Table 3: A — cents of input purchased per dollar of output

	Perf. Arts	Film/Video	Publishing
Perf. Arts	0.10	0.20	0.05
Film/Video	0.15	0.05	0.10
Publishing	0.05	0.10	0.15

Table 4: Value-added ratios and labour coefficients

Industry	GDP per \$ output (v)	Jobs per \$1M output (e)
Perf. Arts	0.55	12.0
Film/Video	0.42	8.5
Publishing	0.48	9.8

Table 5: Leontief Inverse L

	Perf. Arts	Film/Video	Publishing
Perf. Arts	1.1589	0.2543	0.0981
Film/Video	0.1926	1.1081	0.1417
Publishing	0.0908	0.1453	1.1989

Table 6: GDP and Jobs multipliers (per dollar / per \$1M of expenditure)

Industry	GDP Multiplier	Jobs Multiplier
Perf. Arts	0.7619	16.43
Film/Video	0.6750	13.89
Publishing	0.6889	14.13

The Problem

A arts organization spends $D = \$1,000,000$. They have properties that are split 60% Performing Arts and 40% Film/Video. What are its GDP and jobs impacts?

Expenditure: \$ 1e+06

60% Performing Arts | 40% Film/Video

Method 1 — Blend Multipliers Externally

Weight the published multipliers, apply once:

$$m_{\text{blend}}^{GDP} = \sum_k w_k m_k^{GDP} \quad m_{\text{blend}}^{Jobs} = \sum_k w_k m_k^{Jobs}$$

where m_{blend}^{GDP} and m_{blend}^{Jobs} are the organization's weighted-blend GDP and jobs multipliers.

Blended GDP multiplier: 0.7271

Blended Jobs multiplier: 15.42

GDP impact: \$727,142.60

Jobs impact: 15.42

Method 2 — Embed a Synthetic Industry in the Model

Add a 4th column to **A** representing the organization as its own industry. Its input structure (who it buys from) is the 60/40 weighted average of Performing Arts and Film/Video columns. Its row is zeros — no other industry purchases from this composite entity. In small case analysis, this is a reasonable assumption.

Then solve the full 4×4 Leontief system with the same value-added ratios and labour coefficients, and read off the synthetic industry's GDP and jobs multipliers.

Table 7: Expanded A matrix with synthetic industry (4th column)

	Perf. Arts	Film/Video	Publishing	Org (synth)
Perf. Arts	0.10	0.20	0.05	0.14
Film/Video	0.15	0.05	0.10	0.11
Publishing	0.05	0.10	0.15	0.07
Org (synth)	0.00	0.00	0.00	0.00

Synthetic GDP multiplier: 0.7271

Synthetic Jobs multiplier: 15.42

GDP impact: \$727,142.60

Jobs impact: 15.42

Comparison

Table 8: External blend vs. synthetic industry embedded in the model

Metric	External_Blend	Synthetic_Industry	Difference
GDP Multiplier	7.271430e-01	7.271430e-01	0
Jobs Multiplier	1.541830e+01	1.541830e+01	0
GDP Impact (\$)	7.271426e+05	7.271426e+05	0
Jobs Impact	1.541830e+01	1.541830e+01	0

Identical across both GDP and jobs. Blending multipliers externally is algebraically equivalent to running the organization through the I-O model as its own composite industry.

Assumptions

Core I-O model assumptions (apply throughout):

Assumption	What it means
Fixed technical coefficients	Each industry always spends the same share of its output buying inputs from every other industry, regardless of price or scale
Constant returns to scale	Multipliers do not change with shock size; doubling spending doubles impact
No supply constraints	Any industry can expand output to meet demand; no bottlenecks
No price effects	No inflation, wage pressure, or crowding out from the shock itself
Fixed import share	Imports are a constant share of inputs per industry
Homogeneous output	Every dollar of an industry's output has the same input structure

Additional assumptions introduced by mixing:

Assumption	Role
Zero feedback row	The composite organization does not supply other industries; it acts as a final-demand actor
Weighted blend of parents	The organization's input structure is a weighted average of its parent industries' input structures, using chosen weights (output shares)

Mixing inherits the core I-O assumptions above and adds the mixing-specific ones.

CSA Splits - The Least Wrong Way

The Problem

Stats Canada publishes GDP and jobs multipliers (direct, indirect, induced) per IOIC industry. Cultural sub-domains have no published multipliers. Each sub-domain maps to multiple parent IOIC industries via

the CSA methodology.

Where the Three Components Come From

Direct — from CSA data

The CSA publishes GDP, output, and jobs per sub-domain.

- **GDP direct** = GDP / output (both in thousands \$).
- **Jobs direct** = mean quarterly jobs / (annual output / 1000).

Indirect — from parent industries

Each sub-domain's parent IOIC industries have published indirect multipliers. We weight by SUT output (industry size):

$$m_d^{ind} = \frac{\sum_{i \in C_d} \text{output}_i \cdot m_i^{ind}}{\sum_{i \in C_d} \text{output}_i}$$

where d indexes sub-domains, C_d is the set of constituent IOIC industries assigned to sub-domain d , output_i is the SUT gross output of industry i , and m_i^{ind} is the published indirect multiplier for industry i .

The true expenditure split across parents is unobserved. SUT output is the best available proxy.

Induced — ratio adjustment

Baseline induced comes the same way (SUT-weighted from parents). We note:

$$r = \frac{m_d^{indu, baseline}}{m_d^{dir, baseline} + m_d^{ind}}$$

where r is the induced-to-(direct+indirect) ratio, and the *baseline* superscript denotes the SUT output-weighted version computed from constituent IOIC multipliers.

When CSA data replaces the direct, induced updates:

$$m_d^{indu} = r \times (m_d^{dir, CSA} + m_d^{ind})$$

where $m_d^{dir, CSA}$ is the CSA-derived direct multiplier (GDP/output for the sub-domain) that replaces $m_d^{dir, baseline}$.

Same logic for both GDP and jobs.

Obviously this process is repeated independently for each sub-domain and parent domain. Parent domains (e.g. *Visual and applied arts*) are assigned the union of their children's IOIC codes, deduplicated, and treated as a single composite domain where the same SUT output-weighted method applies.

Numerical Example

Table 11: Parent industries: SUT output and published multipliers

Industry	Output	GDP dir	GDP ind	GDP indu	Jobs dir	Jobs ind	Jobs indu
Plastics	800	0.38	0.21	0.13	6.0	3.2	2.0
Textiles	500	0.52	0.14	0.17	11.0	2.1	2.8
Wholesale	700	0.44	0.18	0.15	8.5	2.8	2.3

Table 12: SUT output-based weights

Industry	Output	SUT weight
Plastics	800	0.40
Textiles	500	0.25
Wholesale	700	0.35

Induced ratio r (GDP): 0.2379

Induced ratio r (Jobs): 0.2113

CSA direct GDP (GDP/output): 0.48

CSA direct Jobs (jobs per \$1M): 9.8

Table 13: Baseline vs. CSA-adjusted multipliers

Component	GDP_base	GDP_final	Jobs_base	Jobs_final
Direct	0.436	0.4800	8.12	9.80
Indirect	0.182	0.1820	2.79	2.79
Induced	0.147	0.1575	2.30	2.66
Total	0.765	0.8195	13.21	15.24

CSA data replaced both directs. Induced adjusted via ratio. Indirect unchanged. Each sub-domain is independent.

Induced Must Adjust When Direct Changes

Verified against a closed Leontief model (households endogenous).

Table 14: Open vs closed multipliers — induced ratio is approximately constant

Industry	Open (dir+ind)	Closed (dir+ind+indu)	Induced	Ratio r
Ind 1	0.5683	1.0408	0.4725	0.8314
Ind 2	0.7087	1.2915	0.5828	0.8224
Ind 3	0.6594	1.1873	0.5279	0.8006

Approximately constant — confirming the proportional assumption.

Table 15: Verification: ratio method tracks the model when direct changes

Method	Induced impact
Model induced (closed - open)	0.6025
Ratio method ($r \times$ new open)	0.6024
No adjustment (baseline induced)	0.5250

Ratio method tracks the model. $\square$

Note that obviously in reality we are repeating this for each industry/geography/year etc.

Assumptions: SUT output proportional-draw for indirect weights (true split unobserved); induced/(dir+ind) ratio is constant when direct changes (approximately true in the closed model); each sub-domain is independent.

Final Notes

Full Methodology

For our full code/methodology please see (<https://github.com/SimonRaymond2003/MassCultureMultipliers>).

What Can We Customize?

Any way of measure we use needs to be reproducible for other industries. Thus we can split/mix multipliers etc. and use the IO model. However it is dangerous to over customize our methods to the arts and culture sector. For example what if we said “well artist jobs are different then other sectors because of X and thus we need to adjust things with X methodology”.

This is great for internal use but we need to be able to compare our industry to other sectors/industries. What we do for one industry we need to do for another. Even if we can employ a methodology to both industries we are making the rulebook/types of adjustments to make tailored to our industry. Which is a unfair “advantage” and will be criticized.

Within vs Out of province

The in-province multipliers capture the impact (value-added GDP) within the province while the all-provinces multipliers capture that impact + value added GDP outside of province. This leaking comes only from the indirect and induced impacts as organizations purchase inputs from out-of-province companies.

They are calculated by creating two separate models. The first includes only within province transactions. The second is a model that includes all transactions (within and out of province). The difference between the two models gives us the leakage.

Basic vs Market prices

Basic prices — what the producer receives, excluding taxes on products but including subsidies.

Market prices — what the purchaser pays, including taxes on products and excluding subsidies.

Market prices inflate numbers in high-tax industries. For value-added GDP it is important to use basic prices by definition (European Commission, IMF, OECD, United Nations, and World Bank, 2009).

Are jobs the same thing as employment?

No. “Jobs” is different from employment, here the same person can have multiple jobs. However that individual would be employed only one time (as employment is more of a status).

City-level / Census Division method.

We use Census employment shares (updated every five years) at the Census Division level. Direct impact is assumed to stay 100% in the city as the organization’s own wages and value added happen where it operates. Indirect and induced impacts are scaled by the city’s share of total provincial employment, on the assumption that a city’s supply chain and consumer spending roughly mirrors its share of the provincial labour market.

Formula: City GDP = Direct $\times$ 100% + (Indirect + Induced) $\times$ City Employment Share

Can I apply current multipliers to historical expenditure data?

Yes, but it is situational and we need to be careful of how it is interpreted. It depends on what level of industry specificity you are using. Remember that the Input-Output model is designed to measure the impact of new spending in the economy (e.g. a grant).

Thus if we apply Direct, Indirect and Induced multiplier to the entire arts sector and claim it as contributed GDP we are making a few key mistakes. First if we did this for every sector we would massively inflate GDP as Indirect and Induced GDP represent some other industries direct GDP (the we helped support). Thus this is more of a accounting exercise and needs to be stated as such as it more shows how interconnected the industry is with other industries. Second, if we apply a multiplier to a already existing group of industries we make the assumption that if we remove these industries the multiplier would not change (the ratios of output to GDP and the technical coefficient matrix could possibly change). This is fine/insignificant in the case of a single organization or a representative slice of the sector, but for a entire sector the multipliers will surely change (in this case cease to exist). In reporting a entire sector the Direct effect is clean of this but the Direct effect (GDP/Output) relates to GDP itself and thus it would be easier to skip multipliers and just give the stated GDP for the sector!

Saying “the entire arts sector contributed \$X GDP” via multipliers implicitly compares to a world where the arts sector doesn’t exist. But in that counterfactual, if the arts (or any) sector did not exist that capital and labour would be transferred elsewhere. If every individual in the arts sector was instantly let go of their jobs then those individuals, while not as marginally efficient, would work in other industries. Them working in other industries would in turn raise their GDPs etc. This story applies to all sectors.

Thus, in our application of multipliers to a subsector of the arts sector we point out that this is a upper bound estimate and that the direct impact should be paid more attention to than the indirect and induced impacts.

Why don’t we use a different type of model?

In economics there are two main types of models, Econometric and General Equilibrium. Econometric models are used for causality and sometimes predictions. They do not apply to our use case. General Equilibrium models can work for our use case and may fix many broken assumptions however they are often complex to users. Thus the industry standard has been to use Input-Output Models as the application of them is incredibly simple (applying multipliers).

Does this model account for the social impacts of the arts and culture sector?

No, the model captures GDP and jobs only. It is important to remember that our sector must be comparable to other sectors. If we over customize we lose that ability. GDP and Jobs are what Statistics Canada tracks and thus it was what we have the data for.

Does the model account for volunteer labour?

No, the multipliers are built from Statistics Canada employment and GDP data, which measure paid economic activity. If we are going to compare ourselves to other industries we must all compete on the same benchmark/footing.

Does the model account for the “underground” market?

No, the model is built entirely on Statistics Canada national accounts data, which by definition only covers reported, formal economic activity. While there is a theoretical argument that our sector has a larger underground market(think dealing in cash) than other sectors we would need an empirical way of adding this. In the case that there was some method for estimating this we again would need to do that for every other sector in order to compare sectors/industries.

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